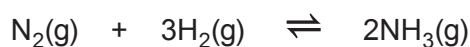


10. (a) Nitrogen can react with hydrogen to form ammonia.



A mixture of  $\text{N}_2$  and  $\text{H}_2$  is left to react at a certain temperature, until it reaches equilibrium. The equilibrium mixture has the following composition.

|               |                             |
|---------------|-----------------------------|
| $\text{N}_2$  | $1.16 \text{ mol dm}^{-3}$  |
| $\text{H}_2$  | $1.60 \text{ mol dm}^{-3}$  |
| $\text{NH}_3$ | $0.752 \text{ mol dm}^{-3}$ |

- (i) A student said that the equilibrium must lie to the left because the concentrations of nitrogen and hydrogen are greater than that of ammonia. Is he correct?

Justify your answer by calculating a value for  $K_c$  for this equilibrium.  
Give the unit for  $K_c$ .

[4]

 $K_c = \dots\dots\dots$ Unit  $\dots\dots\dots$ 

$\dots\dots\dots$   
 $\dots\dots\dots$

- (ii) When the temperature is increased the equilibrium yield of  $\text{NH}_3$  decreases. The student said that the reaction is endothermic. Is he correct?

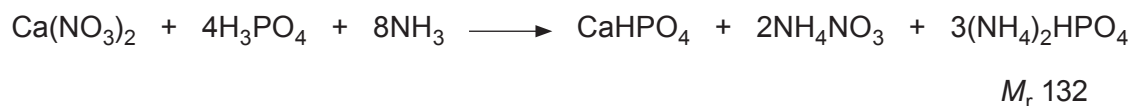
Justify your answer by using Le Chatelier's principle.

[2]

$\dots\dots\dots$   
 $\dots\dots\dots$   
 $\dots\dots\dots$   
 $\dots\dots\dots$



- (b) Ammonia can be used as part of the nitrophosphate process to produce the fertiliser diammonium hydrogenphosphate (DAP) which has the formula  $(\text{NH}_4)_2\text{HPO}_4$ .



Calculate the maximum mass of DAP, in kg, that could be made from 1.00 tonne of ammonia. [3]

Maximum mass = ..... kg

- (c) Calculate the volume, in  $\text{cm}^3$ , that  $2.54 \times 10^{-3}$  mol of nitrogen occupies at a temperature of  $120^\circ\text{C}$  and a pressure of 101 kPa. [4]

Volume = .....  $\text{cm}^3$

